

The Elements of Data Activity: Planning Your Pipeline

You are a data engineer and are assigned to build a pipeline for the following scenario.

Gaming analytics

Your startup company has launched a game that quickly became very popular. The company needs to update the in-game experience to stay competitive. It wants to update designs and rapidly improve gameplay. Currently, the company receives feedback through a feedback button that is embedded in the game navigation and lets users send comments. Gameplay data is sent and received as JSON messages.

Desired outcome: Developers can analyze gaming behaviors in near real time. They can rapidly make design updates that are based on analysis of gameplay data and feedback that was submitted through the feedback button.

Potential data sources

List data sources that might be valuable to meet the desired outcome. For each source, include what you think the format would be. Add as many data sources as you can think of that might be useful, even if you can't fill in details about the data source.

- Think about what types of public datasets might be useful.
- Think about what types of internal or company-owned data might be available.
- Think about any event or time-series data that might be relevant.

Gameplay ⇒ JSON/
Appstore Rating ⇒ JSON/Txt
Public Review ⇒ JSON/Txt
Social Media discussion ⇒ JSON/Txt/API

- ⇒ Public data refer to information available outside company.
- Industry and genre trends
 - App store rating
 - Public Review
 - Social Media Discussion
 - Popularity Report
- Public data helps game team to understand broader market.
- ⇒ Internal or company owned data may be:
- Gameplay event logs
 - Cash Report
 - User information
 - In-app feedback
 - Transaction records
 - Feature used
- This is most accurate and detailed data because it came from game itself.
- ⇒ Event or series data could be:
- Session start and end time
 - Crash timestamp
 - Level completion time
 - Leaderboard update
 - Purchase time
- This data helps to track trends over time.

Value, veracity, and variety considerations

Describe what you need to discover about the sources that you listed to determine if they would be valuable for your use case. Describe the tasks that you think would be needed to get value from the data. Here are some questions to think about:

- What type of analytics and how much processing might be needed to derive insights?
- What type of queries could be made for analysis and visualization?
- What kind of privacy laws might affect data collection and retention across different geographies and industries?
- What kind of security considerations must be addressed? Does the source contain personally identifiable information (PII)?
- What types of veracity challenges might need to be addressed?
- How quickly would the data become stale? • What details do you know about the data (for example, fields and field types)?
- What type of cleaning and transformations might be needed to use this data?

⇒ The system need both real time analytics for quick issue detection and batch analytics for long-term trends.

⇒ Queries may include:

- Completion Rate
- Session Duration
- Crash frequency
- Purchase Trends
- User Activity

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They help us visualize player progress, difficulty, etc.

- ➔ Player data must be protected through encryption, safe storage and secure access control. Only authorized user should access sensitive information.
- ➔ Yes! the data may contain PII such as user IDs and device information. Its issues may include missing logs, duplicate events, etc. Data must be validated and cleaned.
- ➔ Gameplay and feedback data become stale quickly because need fast updates. The data include user ID, event type, source and info. Data such as older record is also useful for performance analysis.
- ➔ Data require validation, duplication and consistent time-stamping. Merging datasets like user profile with event data also improves analysis quality.
- ➔ Firstly, data must be collected, stored securely and check for quality. Then it needs to be cleaned, transformed and combined into useful structures. Then dashboard and analytics must be created.

Velocity and volume considerations

For the data sources that you identified, describe the general requirements for volume and velocity expectations in the context of your selected scenario. Here are some questions to think about:

- What would be the general volume and velocity of incoming data? How might data be transferred into your system (for example, FTP or a connected device)?
- Would the volume of data that is ingested into the pipeline be at steady level, or would it be spiky? Are the patterns predictable?
- Would data be ingested and processed at a regular cadence (for example, nightly or weekly)?
- Does the data need to be processed and analyzed in near real time to be valuable?

- ➔ The game will generate a high volume of events, possibly millions of logs per day depending on active players. Data arrives continuously and at high speed because every player action creates an event. It will be transferred using APIs.
- ➔ The data flow would be spiky, especially during new releases, weekends or special events. Daily pattern are somewhat predictable with higher activity during peak gaming hours.
- ➔ Feedback data and raw gameplay need to be ingested continuously due to real-time requirements. However summaries and reports can be processed on a nightly batch schedule. This create hybrid pipeline.
- ➔ Yes! near real time processing is important for quickly detecting issues like crash or level drop-offs. Real-time analytics help developers responds to problem while players are actively using the game.

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The Elements of Data Activity: Planning Your Pipeline

You are a data engineer and are assigned to build a pipeline for the following scenario.

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E-bike rental proof of concept

Your startup e-bike company has partnered with a local government to pilot an e-bike rental program in several suburban neighborhoods. The government's goal is to reduce carbon emissions and traffic congestion. At this stage, they need to collect data and analyze trends about how the bikes are being used. They also need to determine what data they want to track as they expand the program.

Each of your 50 bikes is equipped with an Internet of Things (IoT) device, which is powered by the bike battery and uses the wireless phone network to transmit data. Devices send and receive JSON data every 2 minutes.

Desired outcome: A proof-of-concept pipeline to collect and analyze data from the IoT devices that are built in to the e-bikes.

Potential data sources
List data sources that might be valuable to meet the desired outcome. For each source, include what you think the format would be. Add as many data sources as you can think of that might be useful, even if you can't fill in details about the data source.
- Think about what types of public datasets might be useful. - Think about what types of internal or company-owned data might be available. - Think about any event or time-series data that might be relevant.
➔ IoT Bike Data:- GPS location, Speed, battery Percentage, timer ➔ User Ride Data:- When trip of data is started/ended, which user number was like ID for the section. Which method is used for payment Physical cash cash / online. ➔ Maintenance Log: → Repair, → Charging, → Battery health ➔ Relevant Event: → Weather Like → Rain, Summer → Roads, Traffic, Population and charging times + station usage

Value, veracity, and variety considerations
Describe what you need to discover about the sources that you listed to determine if they would be valuable for your use case. Describe the tasks that you think would be needed to get value from the data. Here are some questions to think about:
- What type of analytics and how much processing might be needed to derive insights? - What type of queries could be made for analysis and visualization? - What kind of privacy laws might affect data collection and retention across different geographies and industries? - What kind of security considerations must be addressed? Does the source contain personally identifiable information (PII)? - What types of veracity challenges might need to be addressed? - How quickly would the data become stale? • What details do you know about the data (for example, fields and field types)? - What type of cleaning and transformations might be needed to use this data?
• Analytics Needed: → Trip Patterns → route insight → battery usage → maintenance prediction

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- Queries: → Popular routes, → Low batteries alerts, weather correlations
→ busiest time
- Privacy: → GPS, → USER IDS
- Security: Encrypt data, restrict access, monitor pipeline
- Veracity Issues: Missing GPS pointer, sensor errors
network dropouts.
- Freshness: IoT data becomes stale quickly;
weather and maintenance less frequent.
- Cleaning needs: Remove duplicates, handle missing values
→ Standardize time stamps, valid sensor fields.

Velocity and volume considerations

For the data sources that you identified, describe the general requirements for volume and velocity expectations in the context of your selected scenario. Here are some questions to think about:

- What would be the general volume and velocity of incoming data? How might data be transferred into your system (for example, FTP or a connected device)?
- Would the volume of data that is ingested into the pipeline be at steady level, or would it be spiky? Are the patterns predictable?
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- Does the data need to be processed and analyzed in near real time to be valuable?

- IoT bike data arrives in every 2min from each of the 50 bikes
This gives medium-volume data throughout the day.
- Ride events are lower volume but can be spiky during rush hours or weekends.
- Weather, maintenance and demographic data → low velocity and can be scheduled on regular cadence (weekly).
- IoT data requires streaming ingestion and other sources are suitable for batch ingestion that can be processed in daily or weekly batch jobs.

The Elements of Data Activity: Planning Your Pipeline

You are a data engineer and are assigned to build a pipeline for the following scenario.

Fraud detection

You were hired by a company that processes car insurance claims. The company has been losing a lot of money because of fraudulent claims. It is looking for ways to improve the ability to identify fraudulent claims before they are paid. Internal auditors have found that 20 percent of claims that were paid should not have been paid, after underwriting was complete.

The following is a sample of CSV data from the company's claims database:

```
months_as_customer, age, policy_number, policy_bind_date, policy_state, policy_csl, policy_deductable,
policy_annual_premium, umbrella_limit, insured_zip, police_report_available, total_claim_amount, injury_claim,
property_claim, vehicle_claim, auto_make, auto_model, auto_year, fraud_reported
328,48,521585,10/17/2014,OH,250/500,1000,1406.91,0,466132,YES,71610,6510,13020,52080,Saab,92x,2004,Y
228,42,342868,6/27/2006,IN,250/500,2000,1197.22,5000000,468176,?,5070,780,780,3510,Mercedes,E400,2007,Y
134,29,687698,9/6/2000,OH,100/300,2000,1413.14,5000000,430632,NO,34650,7700,3850,23100,Dodge,RAM,2007,N
256,41,227811,5/25/1990,IL,250/500,2000,1415.74,6000000,608117,NO,63400,6340,6340,50720,Chevrolet,Tahoe,2014,Y
228,44,367455,6/6/2014,IL,500/1000,1000,1583.91,6000000,610706,NO,6500,1300,650,4550,Acura,RSX,2009,N
```

Desired outcome: The company has the ability to predict which new claims are fraud with a higher degree of accuracy before they are paid. The stated goal is to reduce the fraudulent claims from 20 percent to 3 percent.

Potential data sources
List data sources that might be valuable to meet the desired outcome. For each source, include what you think the format would be. Add as many data sources as you can think of that might be useful, even if you can't fill in details about the data source. • Think about what types of public datasets might be useful. • Think about what types of internal or company-owned data might be available. • Think about any event or time-series data that might be relevant. we should think about databases like population data, fraud reports about insurance or accidents based statistics. Internal data may come from company databases like claim history, customer profiles, previous fraud cases etc. These can help you to build accurate models. This includes internal reports, employee notes, call records & internal audit findings. This kind of data is unique to the organization & the fraud detection models.

Value, veracity, and variety considerations
Describe what you need to discover about the sources that you listed to determine if they would be valuable for your use case. Describe the tasks that you think would be needed to get value from the data. Here are some questions to think about: • What type of analytics and how much processing might be needed to derive insights? • What type of queries could be made for analysis and visualization? • What kind of privacy laws might affect data collection and retention across different geographies and industries? • What kind of security considerations must be addressed? Does the source contain personally identifiable information (PII)? • What types of veracity challenges might need to be addressed? • How quickly would the data become stale? • What details do you know about the data (for example, fields and field types)? • What type of cleaning and transformations might be needed to use this data?

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- we may need statistical analysis, pattern detection, machine learning or anomaly detection ^{to} to derive insights.
- We might need to clean missing data, ~~run~~ feature engineering & run machine learning models on large datasets.
- Sensitive financial data must be encrypted & all processing must comply with privacy laws.
- Errors may occur from missing, duplicate or incorrect entries; the data contains dates, amount IDs & fraud labels. 14
- We ~~should~~ ^{need to} remove duplicates, fix date format, handle missing values, normalize numbers & create new behavioral features.

Velocity and volume considerations

For the data sources that you identified, describe the general requirements for volume and velocity expectations in the context of your selected scenario. Here are some questions to think about:

- What would be the general volume and velocity of incoming data? How might data be transferred into your system (for example, FTP or a connected device)?
 - Would the volume of data that is ingested into the pipeline be at steady level, or would it be spiky? Are the patterns predictable?
 - Would data be ingested and processed at a regular cadence (for example, nightly or weekly)?
 - Does the data need to be processed and analyzed in near real time to be valuable?
- The data volume will be high because thousands of transactions come in daily & velocity will be continuous.
 - Data will be spiky during peak hours, but overall patterns are predictable.
 - The Fraud related data is usually processed hourly or near real-time, while large 8 files may be processed weekly.
 - ~~As~~ the data needs to be processed in real-time because fraud detection loses value if delayed & fraud activities must be noticed immediately.